1. SALES PERFORMANCE & REVENUE ANALYSIS

- **Total Revenue: What is the total revenue generated in 2015?**

The total revenue generated from pizza sales in 2015 is **₦ 817,860.05**. This represents the total income from all pizza orders during the year.

- **Monthly Revenue Trends**: **How do sales fluctuate across different months? Are there any seasonal patterns?**

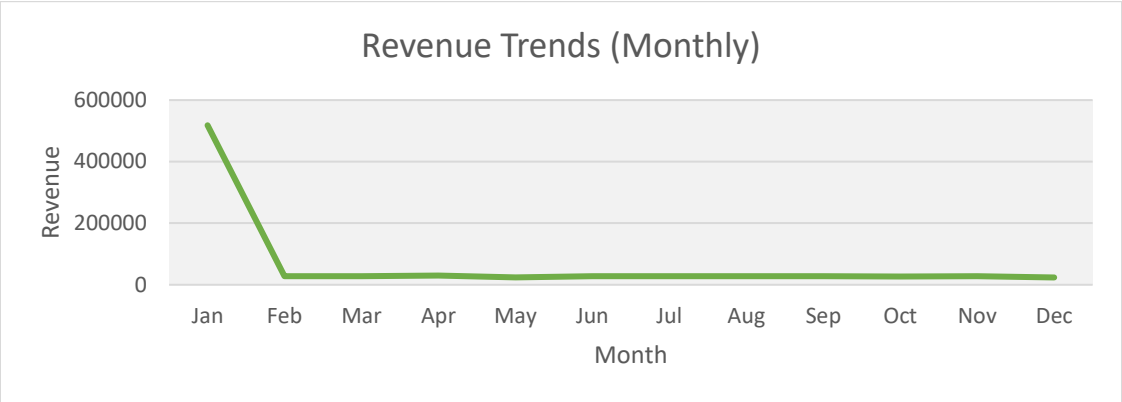


Figure 1: Monthly Revenue Trends

The analysis of monthly revenue reveals how pizza sales fluctuated throughout the year. Revenue appears to increase during certain months and decline in others, indicating possible seasonal patterns. January recorded the highest revenue (**₦ 518,138.80**), significantly outperforming all other months. This sharp peak suggests unusually high customer demand at the beginning of the year, possibly due to festive spending or New Year celebrations.

From February onwards, revenue dropped drastically and stabilized within a relatively narrow range, generally between **₦ 23,000** and **₦ 30,000** per month. This indicates a more consistent but lower level of demand across the remaining months of the year. There are minor variations within this period, with months like April (**₦ 29,675.65**) and September (**₦ 27,963.90**) showing slightly stronger performance, while months such as May (**₦ 23,988.40**) and December (**₦ 24,169.85**) recorded lower revenues.

From the insights of this analysis, a clear conclusion is drawn from the significant spike in January to advice that the business explores marketing, packaging and promotions tailored for seasons like holidays (e.g., New Year, Christmas).

- **Daily & Hourly Sales Distribution: What are the peak sales days and times?**

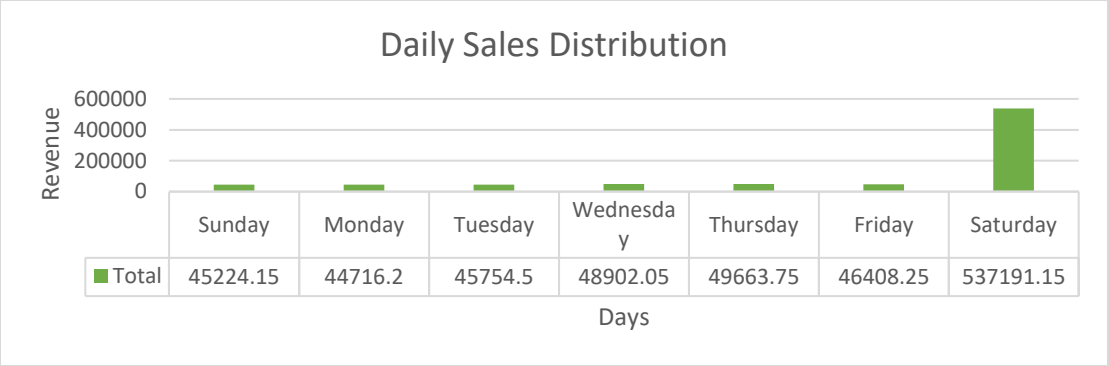


Figure 2: Daily Sales Distribution

The analysis of daily sales shows variations in revenue across different days of the week. Saturday stands out as the highest-performing day, generating approximately (**₦ 537,151**) in revenue, which is substantially higher than all other days. In contrast, sales from Sunday to Friday remain relatively consistent, ranging between approximately **₦ 44,000** and **₦ 49,000**. This indicates stable but moderate customer activity during weekdays.

The sharp increase in revenue on Saturday suggests a strong weekend effect, where customer demand rises significantly. This may be attributed to increased leisure time, social gatherings, and higher ordering behavior during weekends. It is therefore safe to conclude that Saturday is not only the peak sales day, but that the business could experience a significant positive shift in performance by allocating more of its resources, staff, and promotional efforts to this period.

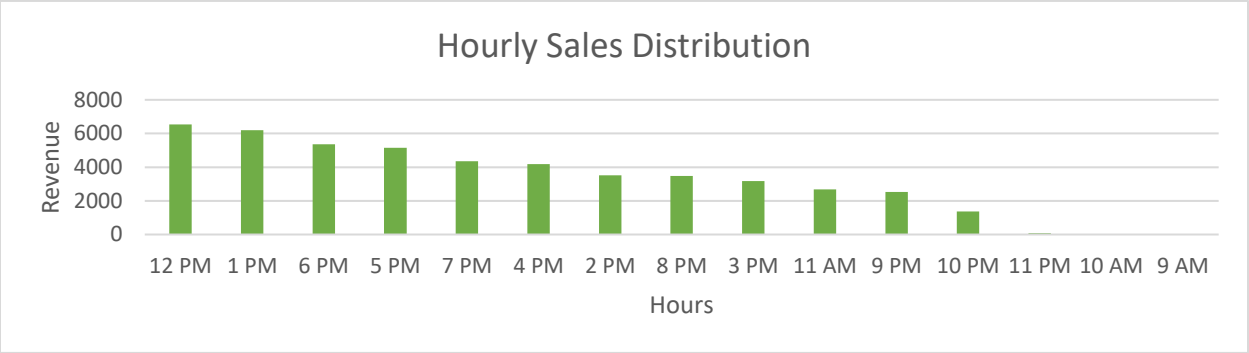


Figure 3: Hourly Sales Distribution

The analysis of hourly sales reveals that customer orders are concentrated around specific periods of the day. The highest number of orders occurs at **12 PM**, with a total of **6,543 orders**, making it the peak hour for pizza sales. This is closely followed by 1 PM, further reinforcing a strong lunchtime demand. In addition to the midday peak, there is another noticeable increase in order activity during the evening hours, particularly between 5 PM and 7 PM. For example, **6 PM** records **5,359 orders**, indicating a secondary peak likely associated with dinner time. Conversely, order volumes are significantly lower during the early morning and late-night hours, with minimal activity recorded around 9 AM and 11 PM. This suggests that customer demand is strongly tied to typical meal times.

It is also important to note that this hourly distribution represents overall customer behavior across all days, and may be influenced by the significantly higher sales observed on peak days such as Saturdays (as seen in Fig. 2 above). In summary, clear peak periods are during lunch and dunner hours of the days of the week, especially Saturdays. Hence, the business can improve efficiency and operations by hiring more staff during these periods, as well as satisfy more customers.

- **Average Order Value (AOV): What is the average revenue per order?**

The Average Order Value (AOV) represents the average amount of revenue generated per customer order. Based on the analysis, the total revenue of **₦ 817,860.05** divided by a total of **48,620 orders** results in an AOV of approximately **₦ 16.82**. This indicates that, on average, each customer order contributes a relatively modest amount to overall revenue. This may suggest that customers typically purchase lower-priced items or small quantities per transaction. Hence, our understanding of AOV tells us that increasing the average order through individual customer-friendly promotions, upselling, or bundled offers could significantly boost overall revenue. Imagine using discounts, combo deals etc. to attract customers into spending more per pizza.

2. PRODUCT PERFORMANCE & MENU OPTIMIZATION

- **Top & Bottom-Selling Pizzas: Which pizza types are the most and least popular based on quantity sold?**

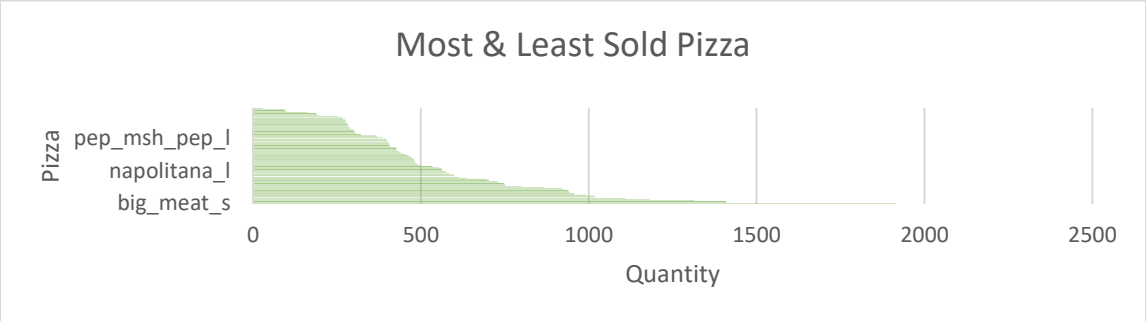


Figure 4: Most & Least Sold Pizza

The analysis of pizza sales based on product popularity reveals clear differences in product performance. The most popular pizza is **big_meat_s**, with a total of **1914 units sold**, and making it the highest-selling item on the menu. Other top-performing pizzas include:

i. **thai_ckn_l** (**1,410 units**)

ii. **five_cheese_l** (**1,409 units**)
iii. **four_cheese_l** (**1,316 units**) and
iv. **classic_dlx_m** (**1,181 units**).

These pizzas are usually demanded most by customers and contribute significantly high to the overall sales volume. Their popularity and customer's choices can be due to either the appealing flavors, familiar ingredients, or fair and inclusive pricing.

On the other hand, the least-selling pizzas include:

i. **the_greek_xxl**
ii. **soppressata_m**
iii. **ckn_pesto_s**
iv. **ckn_pesto_l** and
v. **peppr_salami_m**

These items show limited customer demand, possibly due to higher pricing, less popular ingredient combinations, or niche appeal. In conclusion, adjustments can be made on recipes and promotional budgets for low-performing and especially high-performing pizzas so as to increase its appeal to customers and/or capitalize on rois.

- **Revenue by Pizza Category: How does sales performance vary across different categories (e.g., Vegetarian, Non-Vegetarian, Vegan)?**

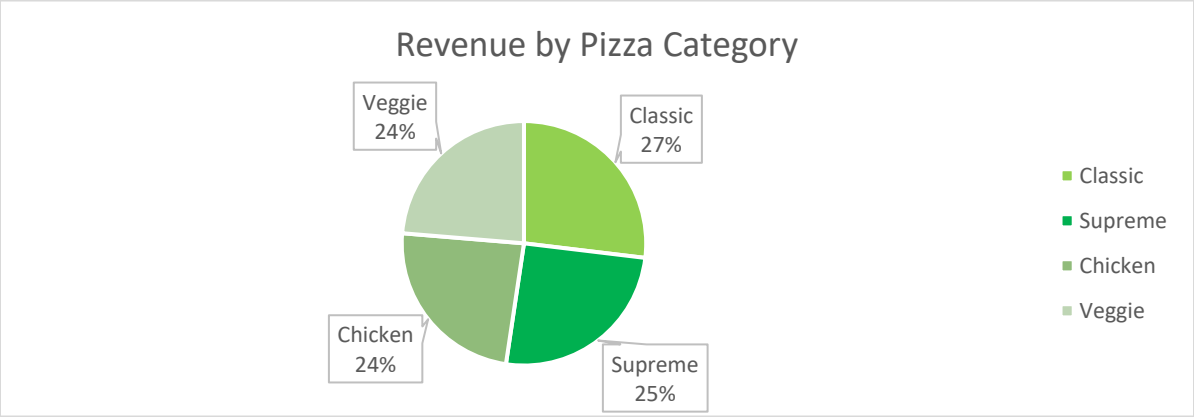


Figure 5: Revenue by Pizza Category

The analysis of revenue by pizza category shows that all categories contribute significantly to total sales, with only slight variations in performance. The **Classic** category generates the highest revenue, contributing approximately **₦ 220,053.10**, which represents about 27% of total revenue. The **Supreme** category follows closely with **₦ 208,197** (25%), while the **Chicken** and **Veggie** categories contribute **₦ 195,919.50** (24%) and **₦ 193,690.45** (24%) respectively. The relatively balanced distribution of revenue across all categories indicates that customer demand is evenly spread, with no single category overwhelmingly dominating sales. However, the slightly higher performance of the **Classic** category suggests a strong preference for traditional and familiar pizza options. In summary, the business has a well-balanced menu and if there is a need to generate more revenue, emphasis could be placed on promoting higher-performing categories like Classic.

- **Revenue by Pizza Size: What proportion of sales comes from small, medium, and large pizzas?**

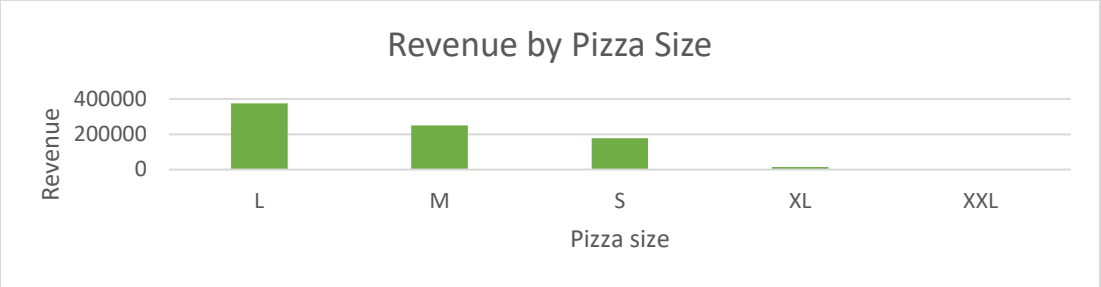


Figure 6: Revenue by Pizza Size

The sales distribution shows a clear preference for the three standard sizes, with a significant drop-off for specialty larger sizes. Top-performing pizza sizes include:

i. Large (L) - **₦ 375,318.70** (approx. 46% of total revenue)
ii. Medium (M) - **₦ 249,382.25**
iii. Small (S) - **₦ 178,076.50**

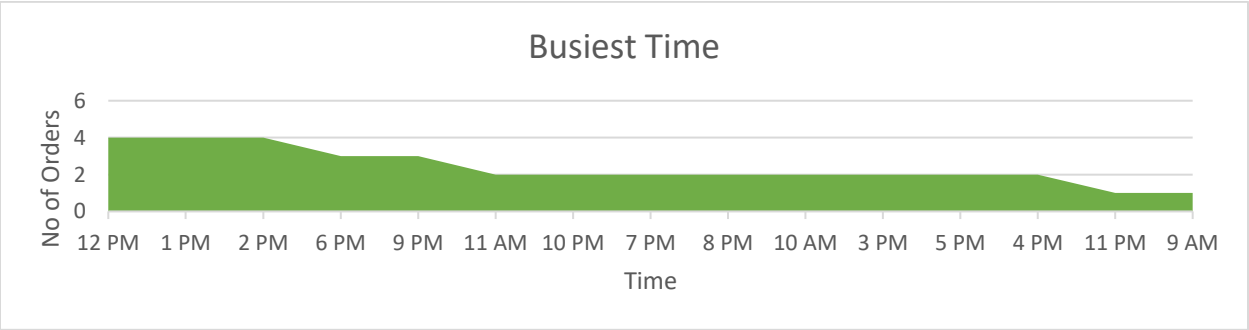
Meanwhile, there seems to be a low sales rates of XL and XXL pizza sizes with XL and XXL contributing **₦ 14,076.00** and **₦ 1,006.60** respectively. In summary, the insights show us that XL and XXL are not cost-effective to promote compared to the L, M and S sizes.

3. CUSTOMER & ORDER BEHAVIOUR

- **Average Quantity per Order: How many pizzas are typically ordered in a transaction?**

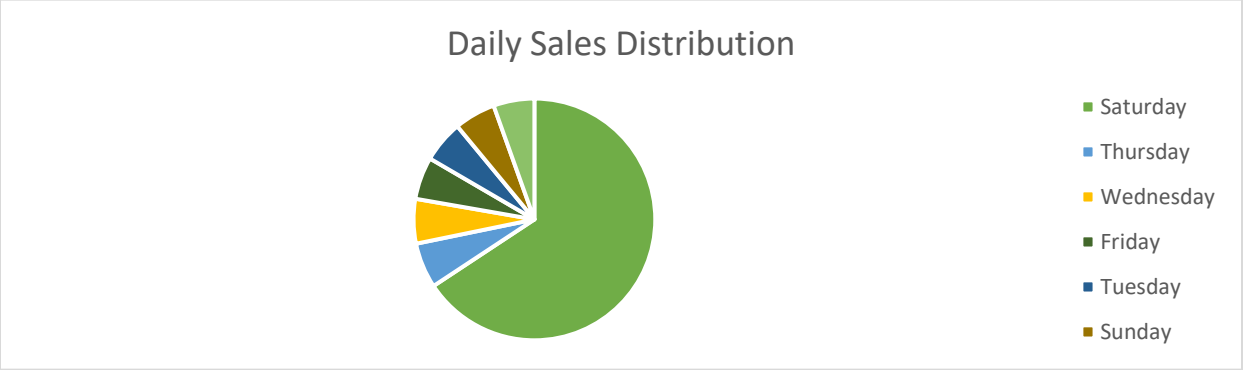
The average quantity of order measures the typical number of pizzas bought in a single transaction. Based on the analysis of total orders bought on hourly basis in comparison with total quantity of pizza sold in terms of quantity, the total quantity of pizzas sold divided by the total number of orders (hourly) results in an average of **0.97 (approximately 1.0)** pizzas per transaction. This could mean that most customers tend to order at least one per transaction. It can be deduced further that there are more individual customers ordering than groups/families and it's suggested that promotions and discounts are made to attract more customers, precisely family discounts/ groups of friends.

- **Most Common Order Time: At what time of the day are most orders placed?**



The analysis shows that the busiest time during the day is **12pm** with at least 4 orders. The analysis further shows that 9am is usually one of the slowest times of day when only 1 orders max are placed by customers. The chart above shows that the company can actually predict when to hire more kitchen staff, delivery staffs, etc. to maximize efficiency, productivity, and build a sustainable brand in which no single arm of the business is wearing out. It helps the business stay grounded on what is working and correct what isn't working. The implication of this is that they make the most money within the hours of 12 PM to 6PM.

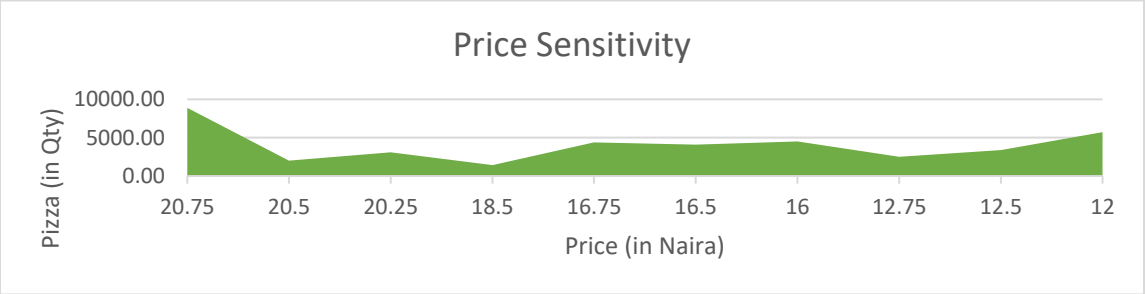
- **Day of the Week Analysis: Which days have the highest and lowest sales volumes?**



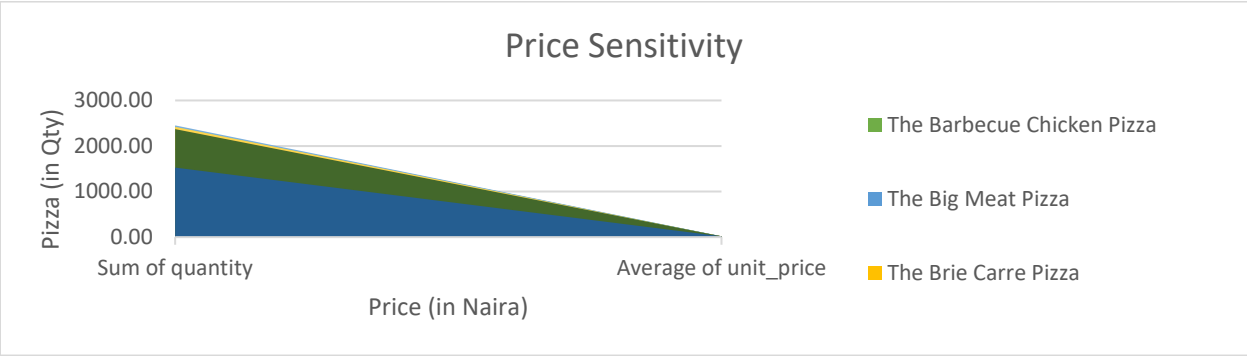
The big slice **(32578, 66%)** is the highest sales volume which happens to fall under **Saturday** by a massive margin, indicating that more than half of our total weekly volume is accounted for during the weekends. this further reveals that a significant majority of customers are most active on weekends. So, while the world moves fast during the week, the weekend particularly Saturday is the most productive for the business. In contrast, the day with the lowest sales is **Monday (2714, 5%)**. This is not surprising since it is the first day of the week for workers immediately after the weekend. If the company is looking to boost sales during the week, a good place to start would be a "Too Busy on Monday to try the new Monday Specials combo?" which could appeal to the busy folks that you recognize their busy schedules but also attract them to an opportunity of a lifetime to unwind.

4. PRICING & DISCOUNT STRATEGY

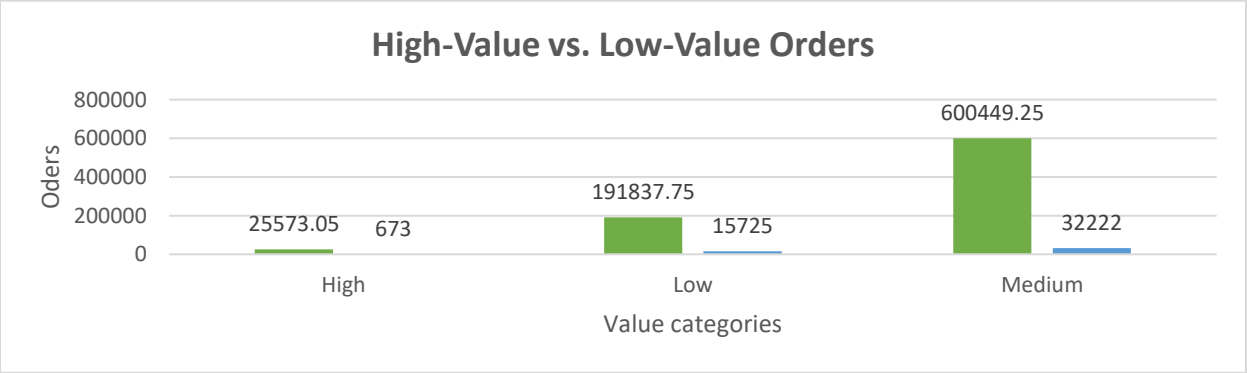
- **Price Sensitivity**: Do higher-priced pizzas sell less compared to lower-priced ones?



The findings reveal that **yes, higher-priced pizzas do sell less compared to lower-priced ones**. There is a downward slope which confirms that as prices increase, the quantity of pizzas sold decreases. For instance, **8891 units (nearly 9,000 units)** happens to have been sold at a unit price of **₦ 20.75**. Although, it's the best-performing in terms of sales volume, there is a drastic drop to prices **₦ 20.5**, **₦ 20.25** and **₦ 18.5** which have pizza sales in quantities way lower than best-selling (**₦ 20.75**). Also, the chart also reveals that the lowest in terms of price vs quantity is at **₦ 18. 5** and which **₦ 20.5** with sales volume of about **1409.00 units to 2026.00 units**.



- **High-Value vs. Low-Value Orders**: What percentage of orders contribute the most revenue?



Analysis revealed that medium-value orders contribute the most share of the total revenue, generating approximately **₦ 600,449.25** from 32,222 orders. Low-value orders also occur frequently, and contribute **₦ 191,837.75** from 15,725 orders. In contrast, high-value orders are relatively rare, accounting for only 673 orders and contributing **₦ 25,573.05** to total revenue. These results indicate that the business is largely driven by customers who make moderate purchases, rather than big spenders or once-in-a-while customers. Hence, the business does not depend on daily sales nor does it depend on its 10% rich customers.

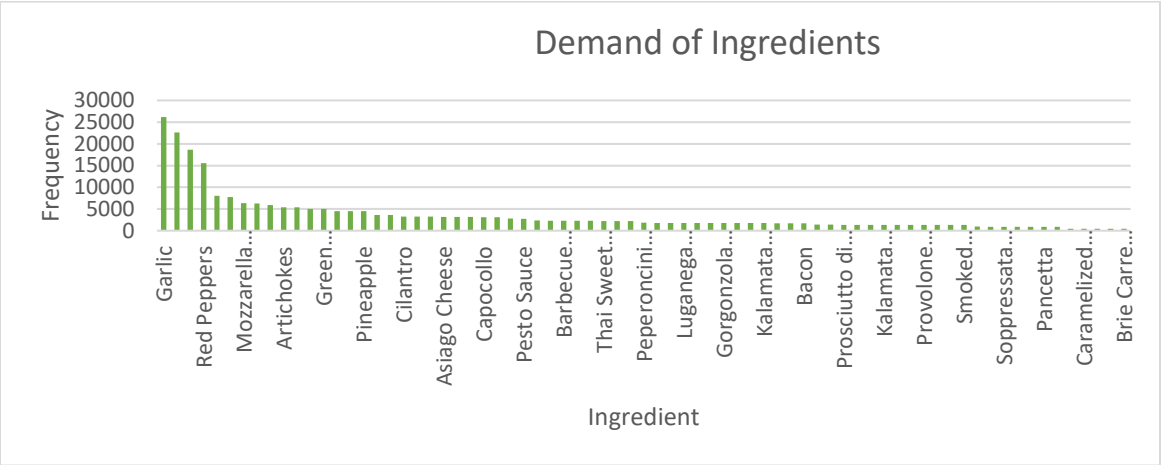
- **Seasonal Pricing Effects**: Are there opportunities for seasonal promotions based on demand trends?

Fig 1: monthly revenue trends reveal noticeable variations in sales performance across the year. January records a significantly higher level of revenue compared to other months, indicating a

strong seasonal spike in demand. This increase can be due to festive activities, holiday celebrations, and increased customer spending during the New Year period. In contrast, other months show moderate spending behavior of customers indicating that there may be seasonality in customer purchasing behavior. Hence, the business can explore seasonal pricing strategies like discounts and offers to stimulate demand and better understand their customers' behavior year long.

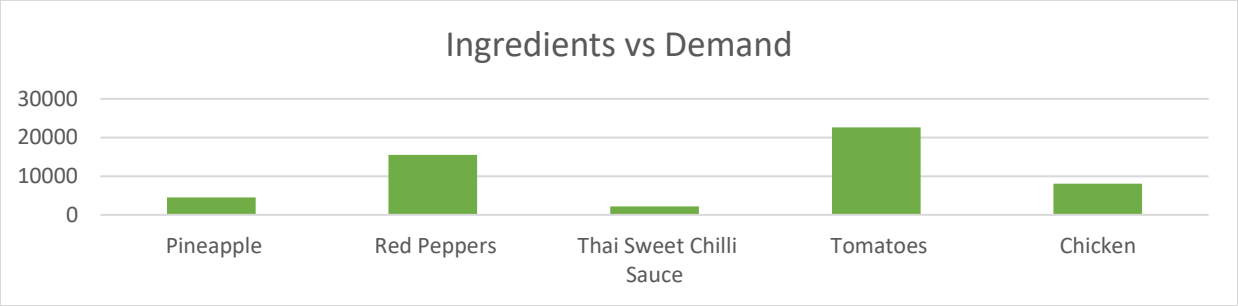
5. INGREDIENT & INVENTORY OPTIMIZATION

- **Most Used Ingredients**: Which ingredients are in highest demand based on sales?



The analysis of ingredients usage reveals clear patterns in customer preferences and menu composition. The most frequently used ingredients include garlic (26,200), tomatoes (22,628), red onions (18,684), and red peppers (15,524), indicating that these ingredients fare at the foundation of many pizza offerings. Other commonly used ingredients include chicken, mushrooms, mozzarella cheese, pepperoni, and green olives. These insights reveal a prediction of investment for the future if such business is seeking to forecast what ingredients to invest in growing themselves in order to cut operating charges. In contrast, ingredients like brie carre cheese, prosciutto, thyme, caramelized onions, and pears recorded the lowest usage levels, each appearing only 458 times, indicating lowest demands. The low usage means that the company can avoid overstocking, and focus future budgets on higher demanded ingredients.

- **Cost vs. Revenue Analysis**: How do ingredient costs compare to revenue generated per pizza type?



The Thai Chicken Pizza generates the highest revenue (**₦ 43,434.25**), making it the top-performing pizza type. An analysis of its ingredients (i.e., Chicken, Pineapple, Tomatoes, Red Peppers, Thai Sweet Chilli Sauce) reveals a combination of both commonly used and specialized ingredients. Key components such as tomatoes, red peppers, and chicken are among the more frequently used ingredients across the dataset, suggesting that they are likely sourced in bulk and at relatively lower costs.

In contrast, ingredients like pineapples, thai sweet chilli sauce adds uniqueness and enhances customer appeal but are the least common and are very minimal in terms of how they affect revenue. It however shows that they are usually used when demanded considering the effect of the pizza type on revenue.

6. OPERATIONAL EFFICIENCY & ORDER FULFILLMENT

- **Order Processing Time**: If timestamps are available, what is the average time between order placement and fulfillment?

No data on delivery time, order time, processing time etc.

- **Delivery vs. Dine-in Sales (if applicable)**: How do sales compare across different service channels?

No data on delivery sales or dine-in sales.